CO 1 ASSESMENT TOOL 3

FLOW CHART ALGORITHM

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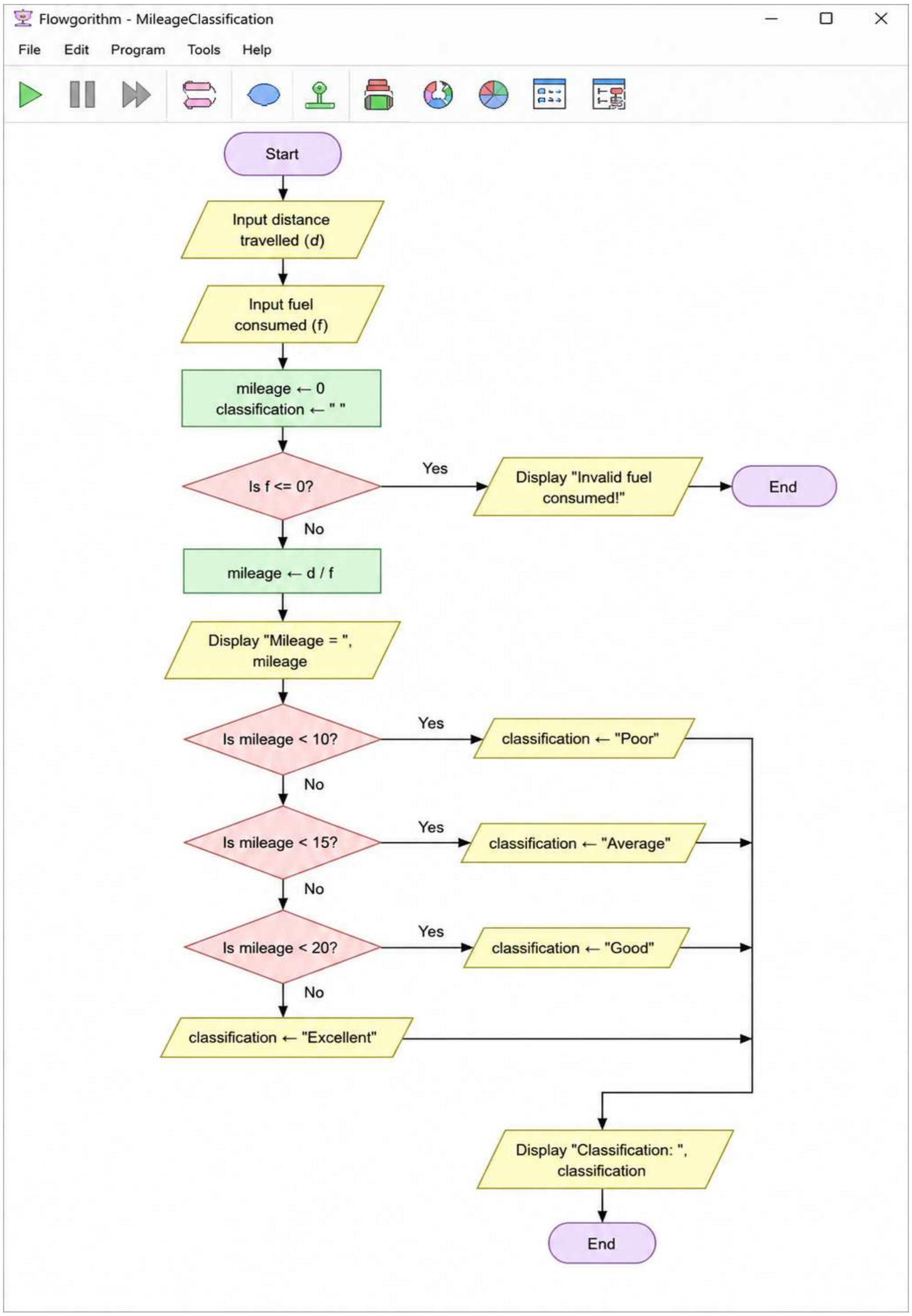
COURSE CODE: CSA0924

COURSE NAME: JAVA PROGRAMMING

SLOT: D

1. **Read distance travelled and fuel consumed. Calculate mileage and classify it as Poor, Average, Good, or Excellent.**



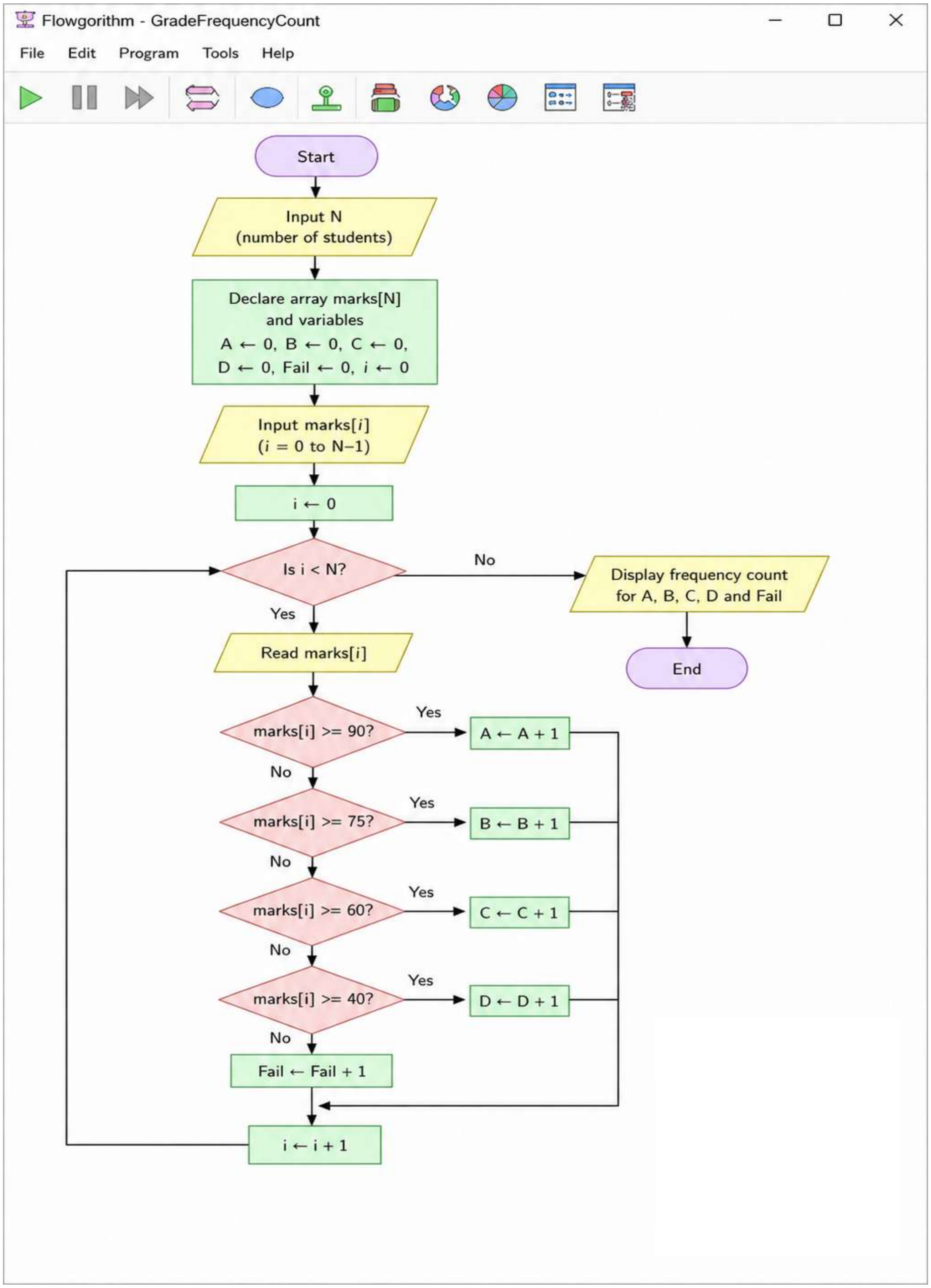
PESUDO CODE:

```
START
Declare Distance, Fuel, Mileage as Real
Input Distance
Input Fuel
IF Fuel <= 0 THEN
   Display "Invalid fuel consumed."
ELSE
   Mileage = Distance / Fuel

   Display "Mileage = ", Mileage

   IF Mileage < 10 THEN
      Display "Poor"
   ELSE IF Mileage < 15 THEN
      Display "Average"
   ELSE IF Mileage < 20 THEN
      Display "Good"
   ELSE
      Display "Excellent"
   END IF
END IF
STOP
```

CODE:

```
import java.util.Scanner;
public class MileageClassification {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        double distance, fuel, mileage;
        System.out.print("Enter distance travelled: ");
        distance = sc.nextDouble();
        System.out.print("Enter fuel consumed: ");
        fuel = sc.nextDouble();
        if (fuel <= 0) {
            System.out.println("Invalid fuel consumed!");
        } else {
            mileage = distance / fuel;
            System.out.printf("Mileage = %.2f%n", mileage);
            if (mileage < 10) {
                System.out.println("Classification: Poor");
            } else if (mileage < 15) {
                System.out.println("Classification: Average");
            } else if (mileage < 20) {
                System.out.println("Classification: Good");
            } else {
                System.out.println("Classification: Excellent");
            }
        }
        sc.close();    }}
```

2. **Read N marks into an array and display a frequency count for grade categories A, B, C, D, and Fail.**



PESUDO CODE:

```
START
DECLARE N, i : INTEGER
DECLARE marks[N] : INTEGER ARRAY
DECLARE A, B, C, D, Fail : INTEGER
READ N
A ← 0
B ← 0
C ← 0
D ← 0
Fail ← 0
FOR i ← 0 TO N - 1 DO
   READ marks[i]

   IF marks[i] >= 90 THEN
      A ← A + 1

   ELSE IF marks[i] >= 75 THEN
      B ← B + 1

   ELSE IF marks[i] >= 60 THEN
      C ← C + 1

   ELSE IF marks[i] >= 40 THEN
      D ← D + 1
   ELSE
```

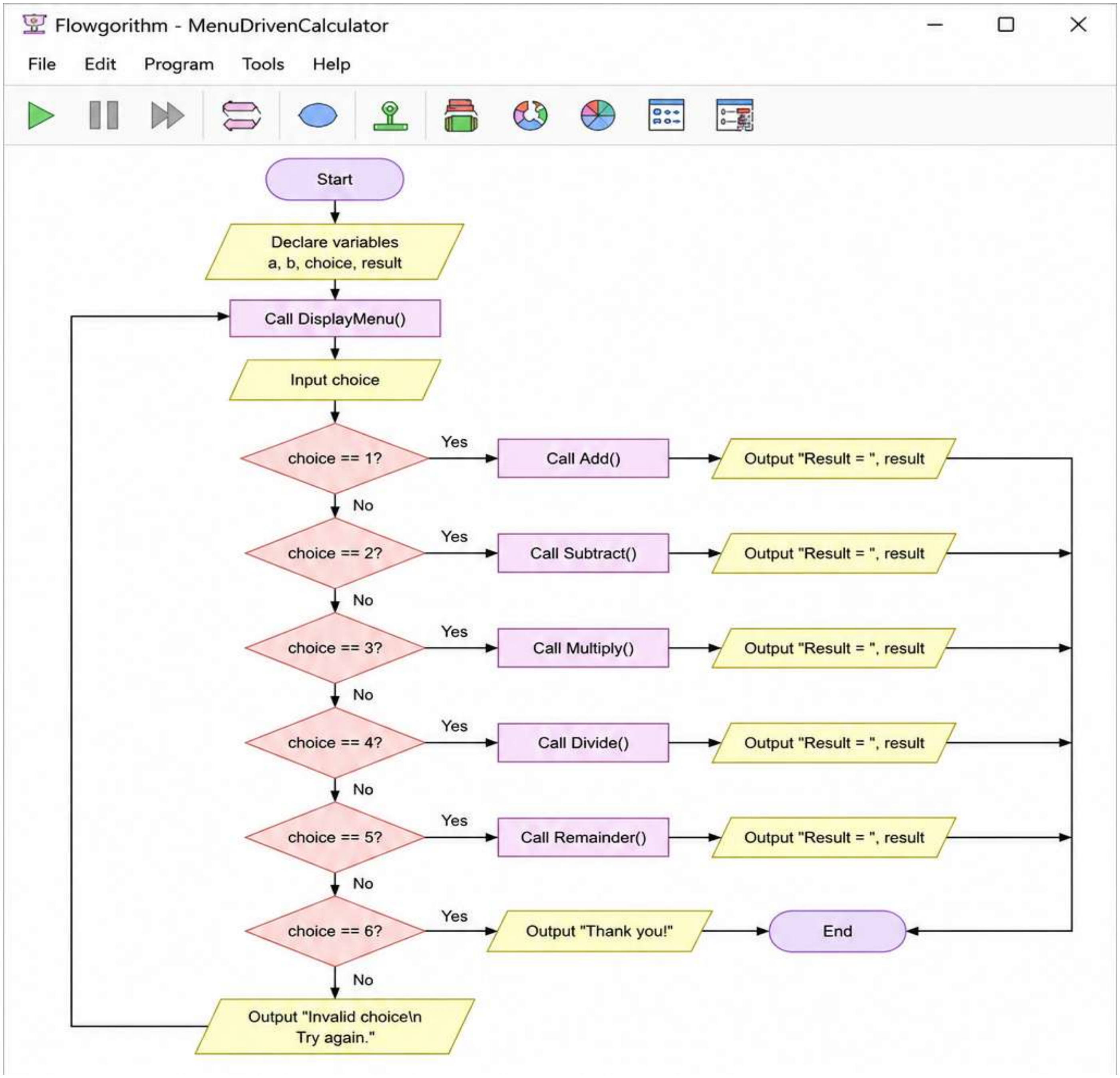
```
        Fail ← Fail + 1

    END IF
END FOR
DISPLAY "Grade Frequency Count"
DISPLAY "A Grade : ", A
DISPLAY "B Grade : ", B
DISPLAY "C Grade : ", C
DISPLAY "D Grade : ", D
DISPLAY "Fail    : ", Fail
STOP
```

JAVA CODE:

```
import java.util.Scanner;
public class GradeFrequencyCount {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the number of students: ");
        int N = sc.nextInt();
        int[] marks = new int[N];
        int A = 0;
        int B = 0;
        int C = 0;
        int D = 0;
        int Fail = 0;
        for (int i = 0; i < N; i++) {
            System.out.print("Enter marks of student " + (i + 1) + ": ");
```

```
            marks[i] = sc.nextInt();
            if (marks[i] >= 90) {
                A++;
            }
            else if (marks[i] >= 75) {
                B++;
            }
            else if (marks[i] >= 60) {
                C++;
            }
            else if (marks[i] >= 40) {
                D++;
            }
            else {
                Fail++;
            }
        }
        System.out.println("\nGrade Frequency Count");
        System.out.println("--------------------------");
        System.out.println("A Grade   : " + A);
        System.out.println("B Grade   : " + B);
        System.out.println("C Grade   : " + C);
        System.out.println("D Grade   : " + D);
        System.out.println("Fail      : " + Fail);
        sc.close();
    }
}
```

3. **Create a menu-driven calculator using separate functions for addition, subtraction, multiplication, division, and remainder**.



PESUDO CODE

```
START
FUNCTION Add(a, b)
   RETURN a + b
END FUNCTION
FUNCTION Subtract(a, b)
   RETURN a - b
END FUNCTION
FUNCTION Multiply(a, b)
   RETURN a * b
END FUNCTION
FUNCTION Divide(a, b)
   IF b = 0 THEN
      DISPLAY "Cannot divide by zero!"
      RETURN 0
   ELSE
      RETURN a / b
   END IF
END FUNCTION

FUNCTION Remainder(a, b)
   IF b = 0 THEN
      DISPLAY "Cannot find remainder by zero!"
      RETURN 0
   ELSE
      RETURN a % b
   END IF
```

```
END FUNCTION
DECLARE choice : INTEGER
DECLARE a, b, result : REAL

DISPLAY "===== MENU ====="
DISPLAY "1. Addition"
DISPLAY "2. Subtraction"
DISPLAY "3. Multiplication"
DISPLAY "4. Division"
DISPLAY "5. Remainder"
DISPLAY "6. Exit"
READ choice
IF choice >= 1 AND choice <= 5 THEN
   READ a
   READ b
END IF
SWITCH(choice)

CASE 1
   result ← Add(a, b)
   DISPLAY "Result = ", result
   BREAK
CASE 2
   result ← Subtract(a, b)
   DISPLAY "Result = ", result
   BREAK
CASE 3
```

```
    result ← Multiply(a, b)
    DISPLAY "Result = ", result
    BREAK
CASE 4
    result ← Divide(a, b)
    DISPLAY "Result = ", result
    BREAK
CASE 5
    result ← Remainder(a, b)
    DISPLAY "Result = ", result
    BREAK
CASE 6
    DISPLAY "Thank You!"
    BREAK
DEFAULT
    DISPLAY "Invalid Choice"
END SWITCH

STOP
```

JAVA CODE:

```java
import java.util.Scanner;
public class MenuDrivenCalculator {
    static double add(double a, double b) {
        return a + b;
    }
    static double subtract(double a, double b) {
```

```
    return a - b;
}
static double multiply(double a, double b) {
    return a * b;
}
static double divide(double a, double b) {
    if (b == 0) {
        System.out.println("Cannot divide by zero!");
        return 0;
    }
    return a / b;
}
static double remainder(double a, double b) {
    if (b == 0) {
        System.out.println("Cannot find remainder by zero!");
        return 0;
    }
    return a % b;
}
public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    int choice;
    double a, b, result;
    System.out.println("===== MENU =====");
    System.out.println("1. Addition");
    System.out.println("2. Subtraction");
    System.out.println("3. Multiplication");
```

```java
System.out.println("4. Division");
System.out.println("5. Remainder");
System.out.println("6. Exit");
System.out.print("Enter your choice: ");
choice = sc.nextInt();
switch (choice) {
    case 1:
        System.out.print("Enter first number: ");
        a = sc.nextDouble();
        System.out.print("Enter second number: ");
        b = sc.nextDouble();
        result = add(a, b);
        System.out.println("Result = " + result);
        break;
    case 2:
        System.out.print("Enter first number: ");
        a = sc.nextDouble();
        System.out.print("Enter second number: ");
        b = sc.nextDouble();
        result = subtract(a, b);
        System.out.println("Result = " + result);
        break;
    case 3:
        System.out.print("Enter first number: ");
        a = sc.nextDouble();
        System.out.print("Enter second number: ");
        b = sc.nextDouble();
```

```
            result = multiply(a, b);
            System.out.println("Result = " + result);
            break;
        case 4:
            System.out.print("Enter first number: ");
            a = sc.nextDouble();
            System.out.print("Enter second number: ");
            b = sc.nextDouble();
            result = divide(a, b);
            System.out.println("Result = " + result);
            break;
        case 5:
            System.out.print("Enter first number: ");
            a = sc.nextDouble();
            System.out.print("Enter second number: ");
            b = sc.nextDouble();
            result = remainder(a, b);
            System.out.println("Result = " + result);
            break;
        case 6:
            System.out.println("Thank You!");
            break;
        default:
            System.out.println("Invalid Choice!");
    }
    sc.close();
}}
```